DSD

Lab Report No 10

DSD lab

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Section: B

"On my honor , as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work"

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Change the functionality of the lock such that it unlocks on the sequence of 11011.

```
//clock_divider
module clock_divider(
    input clk_in,
    output reg clk_out
);
   reg [27:0] counter = 28'd0;
            parameter divisor = 28'd1000000;

    always @(posedge clk_in) begin
                        counter <= counter + 28'd1;

                        if (counter >= (divisor - 1))
            counter <= 28'd0;

                        clk_out <=(counter < divisor/2)?1'b1:1'b0;
                        end
endmodule



//lock_fsm2
module lock_fsm2(
            input btn0,
            input btn1,
            input clk,
```

```
	input RST_BTN,
	output reg led,
	output reg [3:0]bcd
);
	reg [2:0]state;

	parameter s0 = 3'b000;
	parameter s1 = 3'b001;
	parameter s2 = 3'b010;
	parameter s3 = 3'b011;
	parameter s4 = 3'b100;
	parameter s5 = 3'b101;

	always@(posedge clk) begin
		if(RST_BTN)
			state <= s0;

		else
			case(state)
				s0: begin
					if(btn0)
						state <= s0;
					else if(btn1)
						state <= s1;
					else
						state <= s0;
```

```
end

s1: begin
        if(btn0)
                state <= s1;
        else if(btn1)
                state <= s2;
        else
                state <= s1;
end

s2: begin
        if(btn0)
                state <= s3;
        else if(btn1)
                state <= s2;
        else
                state <= s2;
end

s3: begin
        if(btn0)
                state <= s3;
        else if(btn1)
                state <= s4;
        else
```

```
                        state <= s3;
                end

                s4: begin
                        if(btn0)
                                state <= s4;
                        else if(btn1)
                                state <= s5;
                        else
                                state <= s4;
                end

                s5: begin
                        if(btn0)
                                state <= s5;
                        else if(btn1)
                                state <= s0;
                        else
                                state <= s5;
                end
        endcase
end



always@(*)
        case(state)
```

```
s0: begin
        led <= 0;
        bcd <= 4'b0000;
end

s1: begin
        led <= 0;
        bcd <= 4'b0001;
end


s2: begin
        led <= 0;
        bcd <= 4'b0010;
end

s3: begin
        led <= 0;
        bcd <= 4'b0011;
end

s4: begin
        led <= 0;
        bcd <= 4'b0100;
end
```

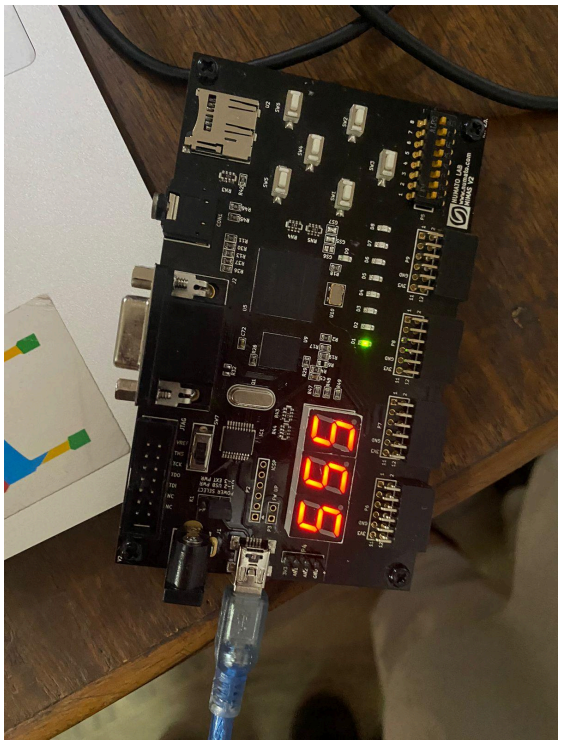
```
                s5: begin
                        led <= 1;
                        bcd <= 4'b0101;
                end
            endcase

endmodule



//synchronizer
module synchronizer(
        input clk,
        input btn,
        input rst,
        output synch_btn
    );

        wire Q1;

        D_FF df1(Q1, btn, clk, rst);
        D_FF df2(synch_btn, Q1, clk, rst);

endmodule

// D_FlipFLop
module D_FF(Q,D,clk,rst);
```

```
input D; // Data input
input clk; // clock input
input rst;
output reg Q; // output Q
always @(posedge clk or negedge clk)
begin
if(rst)
        Q = 1'b0;
else
        Q = D;
end
endmodule



//level_to_pulse
module level_to_pulse(
        input synch_input,
        input clk,
        input rst,
        output pulse
   );
         wire Q;

         D_FF df(Q, synch_input, clk, rst);
         and a(pulse ,~Q, synch_input);
endmodule
```

```
//btn_module

module btn_module(
        input btn,
        input CLK,
        input RST,
        output pulse
   );
        wire synch_btn;
        synchronizer s1(CLK, btn, RST, synch_btn);
        level_to_pulse lp1(synch_btn, CLK, RST, pulse);

endmodule



//BCD_to_SevenSeg

module BCD_to_SevenSeg(
   input [3:0] bcd,
        output reg [7:0] O
        );
```

```
	always @(*)
		case(bcd)	//gfedcba
			4'b0000: O = 8'b11000000;
			4'b0001: O = 8'b11111001;
			4'b0010: O = 8'b10100100;
			4'b0011: O = 8'b10110000;
			4'b0100: O = 8'b10011001;
			4'b0101: O = 8'b10010010;
			4'b0110: O = 8'b10000010;
			4'b0111: O = 8'b11111000;
			4'b1000: O = 8'b10000000;
			4'b1001: O = 8'b10010000;
			default: O = 8'b11111111;
		endcase

endmodule




//Top Module
module Top( CLK, RST, RST_BTN, BTN0, BTN1, LED, SEVENSEG);

	input CLK, RST, BTN0, BTN1, RST_BTN;
	output LED;
```

```
        output [7:0]SEVENSEG;
        wire SLOW_CLOCK;
        //wire synch_btn0, synch_btn1, synch_rst;
        wire pulse0, pulse1, RST_Pulse;
        wire [3:0]bcd;

        clock_divider divider(CLK, SLOW_CLOCK);
        lock_fsm2 my_fsm( pulse0, pulse1, SLOW_CLOCK, RST_Pulse, LED, bcd);
        btn_module b0(BTN0, SLOW_CLOCK, RST,  pulse0);
        btn_module b1(BTN1, SLOW_CLOCK, RST,  pulse1);
        btn_module b2(RST_BTN, SLOW_CLOCK, RST,  RST_Pulse);

    /*
        synchronizer s1(SLOW_CLOCK, BTN0, RST, synch_btn0);
        level_to_pulse lp1(synch_btn0, SLOW_CLOCK, RST, pulse0);
        synchronizer s2(SLOW_CLOCK, BTN1, RST, synch_btn1);
        level_to_pulse lp2(synch_btn1, SLOW_CLOCK, RST, pulse1);
        synchronizer s3(SLOW_CLOCK, RST_BTN, RST, synch_rst);
        level_to_pulse lp3(synch_rst, SLOW_CLOCK, RST, RST_pulse);
        */
        BCD_to_SevenSeg(bcd, SEVENSEG);

endmodule
```



2) Change the functionality of the lock such that it unlocks on the sequence of 01011

```
//clock_divider
module clock_divider(
    input clk_in,
    output reg clk_out
);
    reg [27:0] counter = 28'd0;
    parameter divisor = 28'd1000000;

    always @(posedge clk_in) begin
        counter <= counter + 28'd1;
```

```
        if (counter >= (divisor - 1))
            counter <= 28'd0;

        clk_out <=(counter < divisor/2)?1'b1:1'b0;
        end
endmodule
//lock_fsm2
module lock_fsm2(
    input btn0,
    input btn1,
    input clk,
    input RST_BTN,
    output reg led,
    output reg [3:0]bcd
    );
    reg [4:0]state;  // Changed to 5 states to recognize 01011

    parameter s0 = 5'b00000;
    parameter s1 = 5'b00001;
    parameter s2 = 5'b00010;
    parameter s3 = 5'b00011;
    parameter s4 = 5'b00100;
    parameter s5 = 5'b00101; // Added state s5
    // parameter s6 = 5'b00110; // Not needed for 01011, removed.

    always@(posedge clk) begin
```

```
if(RST_BTN)
    state <= s0;

else
    case(state)
        s0: begin
            if(btn0)
                state <= s1; // 0
            else if(btn1)
                state <= s0;
            else
                state <= s0;
        end

        s1: begin
            if(btn0)
                state <= s1;
            else if(btn1)
                state <= s2; // 1
            else
                state <= s0;
        end

        s2: begin
            if(btn0)
                state <= s3; // 0
```

```
    else if(btn1)
        state <= s0;
    else
        state <= s0;
end

s3: begin
    if(btn0)
        state <= s1;
    else if(btn1)
        state <= s4; // 1
    else
        state <= s0;
end

s4: begin
    if(btn0)
        state <= s5; // 1
    else if (btn1)
        state <= s0;
    else
        state <= s0;
end

s5: begin  // Added state for the final 1
    if(btn0)
```

```
                state <= s1;
            else if(btn1)
                state <= s0;
            else
                state <= s5;
        end
    endcase
end



always@(*)
  case(state)
    s0: begin
        led <= 0;
        bcd <= 4'b0000;
    end

    s1: begin
        led <= 0;
        bcd <= 4'b0001;
    end



    s2: begin
        led <= 0;
        bcd <= 4'b0010;
```

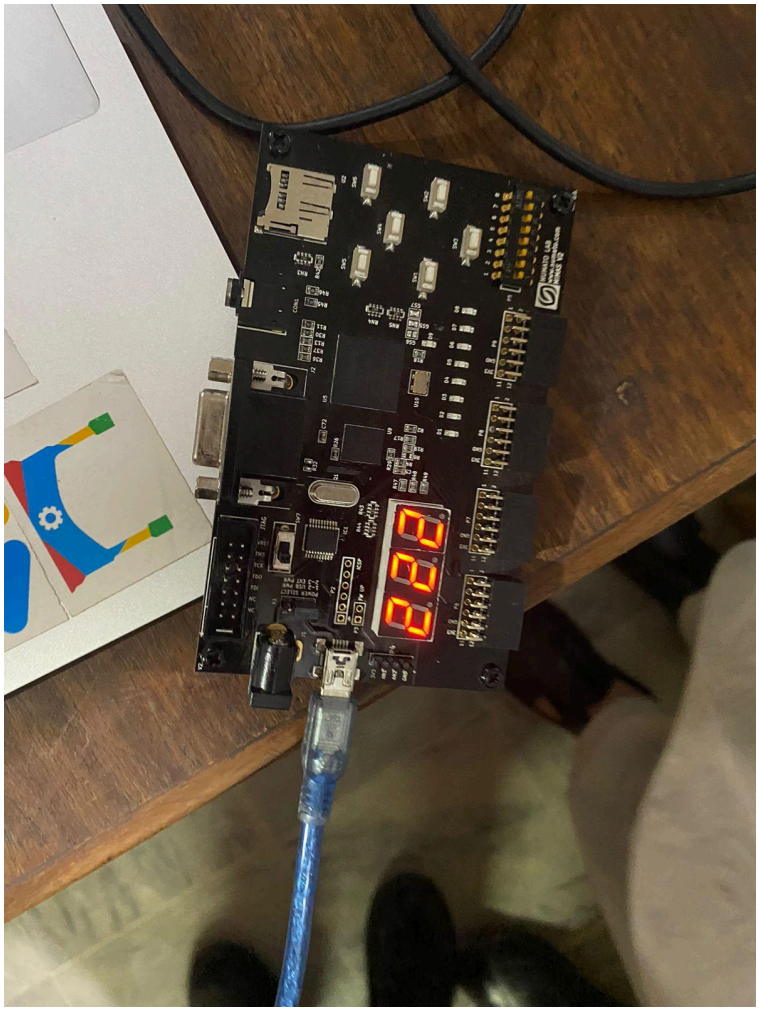
```
            end

            s3: begin
                led <= 0;
                bcd <= 4'b0011;
            end

            s4: begin
                led <= 0;
                bcd <= 4'b0100;
            end

            s5: begin
                led <= 1;  // LED is turned on when the sequence 01011 is detected.
                bcd <= 4'b0101;
            end
        endcase

endmodule



//synchronizer
module synchronizer(
    input clk,
    input btn,
    input rst,
```

```
    output synch_btn
    );

    wire Q1;

    D_FF df1(Q1, btn, clk, rst);
    D_FF df2(synch_btn, Q1, clk, rst);

endmodule

// D_FlipFLop
module D_FF(Q,D,clk,rst);
input D; // Data input
input clk; // clock input
input rst;
output reg Q; // output Q
always @(posedge clk or negedge clk)
begin
if(rst)
    Q = 1'b0;
else
    Q = D;
end
endmodule
```

```
//level_to_pulse
module level_to_pulse(
    input synch_input,
    input clk,
    input rst,
    output pulse
    );
     wire Q;

     D_FF df(Q, synch_input, clk, rst);
     and a(pulse ,~Q, synch_input);
endmodule
```

```
//btn_module

module btn_module(
    input btn,
    input CLK,
    input RST,
    output pulse
    );
    wire synch_btn;
```

```
    synchronizer s1(CLK, btn, RST, synch_btn);
    level_to_pulse lp1(synch_btn, CLK, RST, pulse);

endmodule



//BCD_to_SevenSeg

module BCD_to_SevenSeg(
    input [3:0] bcd,
    output reg [7:0] O
    );
    always @(*)
        case(bcd)        //gfedcba
            4'b0000: O = 8'b11000000;
            4'b0001: O = 8'b11111001;
            4'b0010: O = 8'b10100100;
            4'b0011: O = 8'b10110000;
            4'b0100: O = 8'b10011001;
            4'b0101: O = 8'b10010010;
            4'b0110: O = 8'b10000010;
            4'b0111: O = 8'b11111000;
            4'b1000: O = 8'b10000000;
            4'b1001: O = 8'b10010000;
            default: O = 8'b11111111;
        endcase
```

```
endmodule



//Top Module
module Top( CLK, RST, RST_BTN, BTN0, BTN1, LED, SEVENSEG);

    input CLK, RST, BTN0, BTN1, RST_BTN;
    output LED;
    output [7:0]SEVENSEG;
    wire SLOW_CLOCK;
    //wire synch_btn0, synch_btn1, synch_rst;
    wire pulse0, pulse1, RST_Pulse;
    wire [3:0]bcd;

    clock_divider divider(CLK, SLOW_CLOCK);
    lock_fsm2 my_fsm( pulse0, pulse1, SLOW_CLOCK, RST_Pulse, LED, bcd);
    btn_module b0(BTN0, SLOW_CLOCK, RST,  pulse0);
    btn_module b1(BTN1, SLOW_CLOCK, RST,  pulse1);
    btn_module b2(RST_BTN, SLOW_CLOCK, RST,  RST_Pulse);

    /*
    synchronizer s1(SLOW_CLOCK, BTN0, RST, synch_btn0);
```

```
    level_to_pulse lp1(synch_btn0, SLOW_CLOCK, RST, pulse0);
    synchronizer s2(SLOW_CLOCK, BTN1, RST, synch_btn1);
    level_to_pulse lp2(synch_btn1, SLOW_CLOCK, RST, pulse1);
    synchronizer s3(SLOW_CLOCK, RST_BTN, RST, synch_rst);
    level_to_pulse lp3(synch_rst, SLOW_CLOCK, RST, RST_pulse);
    */
    BCD_to_SevenSeg(bcd, SEVENSEG);

endmodule
```